

Mega7 Simulation Analysis Report

0. Executive summary

Mega7 is the **deepest balance sweep to date** and the first to use the new pooled `ratings_combined.csv` artefact (every battle from every stage and weather in one flat table). It spans **nine team sizes (2v2–10v10)** × **six weathers**, 360 pieces (120 bases × 3 levels), with a combined median of **237 matches/piece** — ~20× the per-cell depth of mega6.

Five headline findings:

- Faction balance is sound.** Champion vs enemy win rate sits at 0.507 / 0.491 pooled and stays within ±0.02 at every team size. Every pathology below is **within** the roster, not a champion/enemy asymmetry.
- The mage deficit persists and is a small-format problem.** Mage is the weakest role pooled (win rate 0.458 vs non-mage 0.515; gap +0.057). The within-tier deficit is worst at 2v2 (0.079), shrinks through the mid sizes, and **inverts** at 7v7–8v8 (mages slightly ahead). Same shape as mega6, marginally worse at 2v2.
- Weather favor is modest and working correctly** — ~+0.01, not the +0.18 a first pass suggested. An earlier read of this dataset reported a dominant +0.18 own-vs-counter swing; that was a **metric artefact** (the `counter_weather_wr` column zero-fills weathers a piece never played — 37% of rows, including every `clear`-affinity piece — and averaging those zeros craters the counter mean). Recomputed from raw per-weather win rates, the own-weather edge is **+0.010 own-vs-counter** and **+0.008 own-vs-clear-baseline**, positive for **every** affinity (no snow/thunder inversion — that too was thin-slice noise). The favor system is applied and behaves as designed; the metric has been fixed (§6).
- T10 hybrid boss kits under-deliver vs their power budget** — **again**. Hybrid has the **highest** raw win rate (0.559) but the **most negative** pooled `wr_delta` (−0.030): Aurion (clear T10 L3, −0.245), Aerion (thunder T10 L3, −0.165), Storm Tyrant (−0.089). They win because they are high-power, but less than their power says they should. This recurs from mega6.
- Level premium is steep but roughly on-budget.** L3 pooled win rate 0.593 vs L1 0.424; L3 `wr_delta` −0.021 (slightly over-delivers). By design, not flagged.

One real watch-item: **timeouts climb to 44% at 10v10**, compressing role signal at the largest sizes.

1. Dataset and method

Field	Value
Directory	<code>results/mega/mega7/</code>
Run date	2026-06-03
Engine state	current working tree: poison percentage-decay (V.25 <code>decay_fraction=0.2</code>), barrier system, Glade Heron rework, weather favor ±0.30
Stages	2v2, 3v3, 4v4, 5v5, 6v6, 7v7, 8v8, 9v9, 10v10 (no 1v1)
Per-stage rated rows	19,200
Combined pieces	360 (120 bases × 3 levels)
Weathers	clear, cloudy, mist, rain, snow, thunder
max-ticks	12,000 (engine default — sudden death engaged)
Baseline	mega6 (<code>cache_mega6.rds</code> ; raw dir deleted)

The combined file is the canonical aggregate. Per-(stage,weather) ratings still exist for slicing, but per-cell

samples are thin (median 3–6 matches). `ratings_combined.csv` pools every battle — pair-game weighted — so its per-piece win rates and cross-weather metrics are the reliable read (median 237 matches). All §7–§8 outlier work uses it.

wr_delta interpretation (carries from prior reports). `expected_wr` is the deterministic power-threshold model: 1.0 if your team’s ΣP exceeds the opponent’s, 0.0 if less, 0.5 if equal, averaged over the actual opponent field. `wr_delta = win_rate - expected_wr`. Near 0 = on-budget; negative = kit under-delivers vs power; positive = over-delivers.

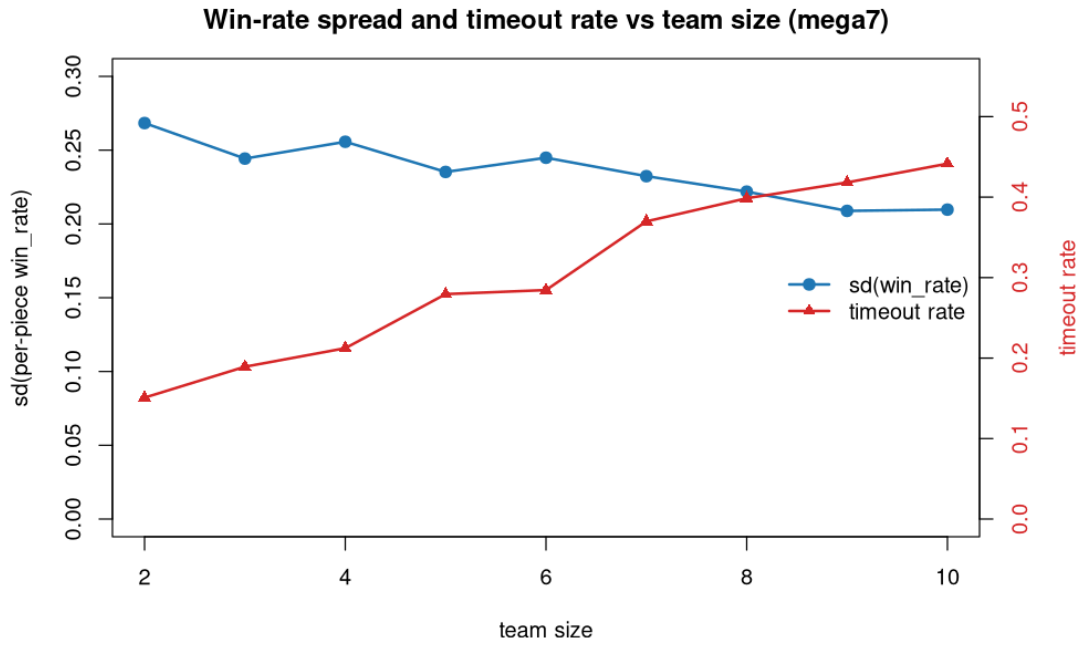
Provenance caveat. Mega7 was generated from the current uncommitted working tree. Kit-level reads for pieces touched this session (Glade Heron, poison users, Hierarchy) are indicative; a clean post-commit re-run is advisable before acting on those specific rows.

2. Aggregate balance

stage	champion	enemy	sd(win_rate)	timeout
2v2	0.500	0.505	0.268	0.151
3v3	0.505	0.492	0.244	0.189
4v4	0.501	0.491	0.256	0.212
5v5	0.489	0.493	0.235	0.279
6v6	0.502	0.503	0.245	0.285
7v7	0.508	0.475	0.232	0.370
8v8	0.519	0.486	0.222	0.399
9v9	0.513	0.489	0.209	0.418
10v10	0.504	0.485	0.210	0.442

Champion/enemy parity holds across all formats. Win-rate spread declines only modestly with team size (0.27 → 0.21), much less than mega6’s collapse (0.23 → 0.09). The cause is **per-stage sampling depth**, not a real effect: mega7’s per-(piece, stage) samples are thin (median 3–6 matches — the `--total-battles` budget spread across nine sizes × six weathers), so each per-piece win rate carries large binomial noise that inflates sd at every size. mega6 reached 50–222 matches/piece at its top sizes, so its sd genuinely collapsed. Read mega7’s **pooled** (combined) win rates for per-piece signal; treat per-stage sd as noise-dominated.

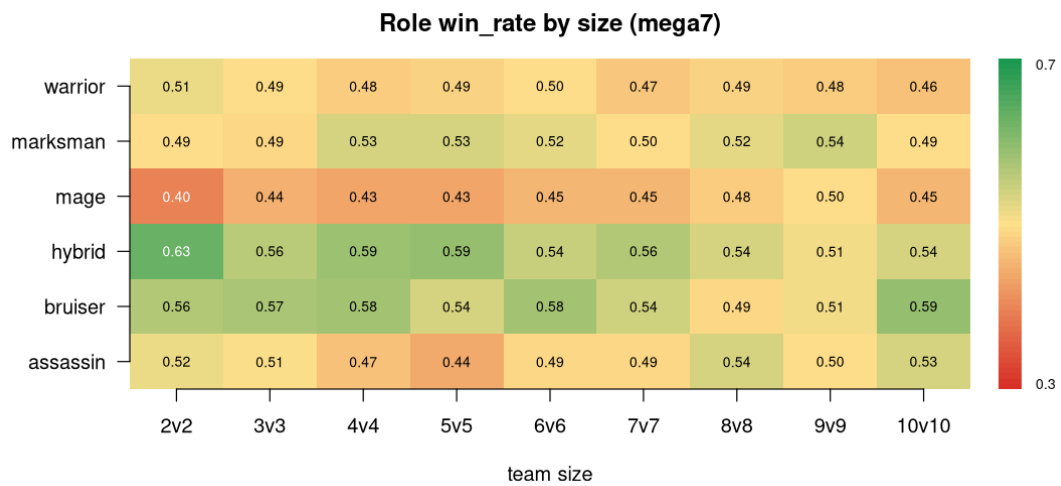
Timeouts are a rising concern: 15% at 2v2 → 44% at 10v10. Every timeout is a draw, pulling win rates toward 0.50 and compressing differences at the top sizes.



win\{}_rate spread and timeout vs team size

3. Role balance

3.1 Win rate by team size



Role win\{}_rate by size (heatmap)

Pooled (combined) role win rates:

role	win_rate
mage	0.458
warrior	0.486
assassin	0.496
marksman	0.515
bruiser	0.546
hybrid	0.559

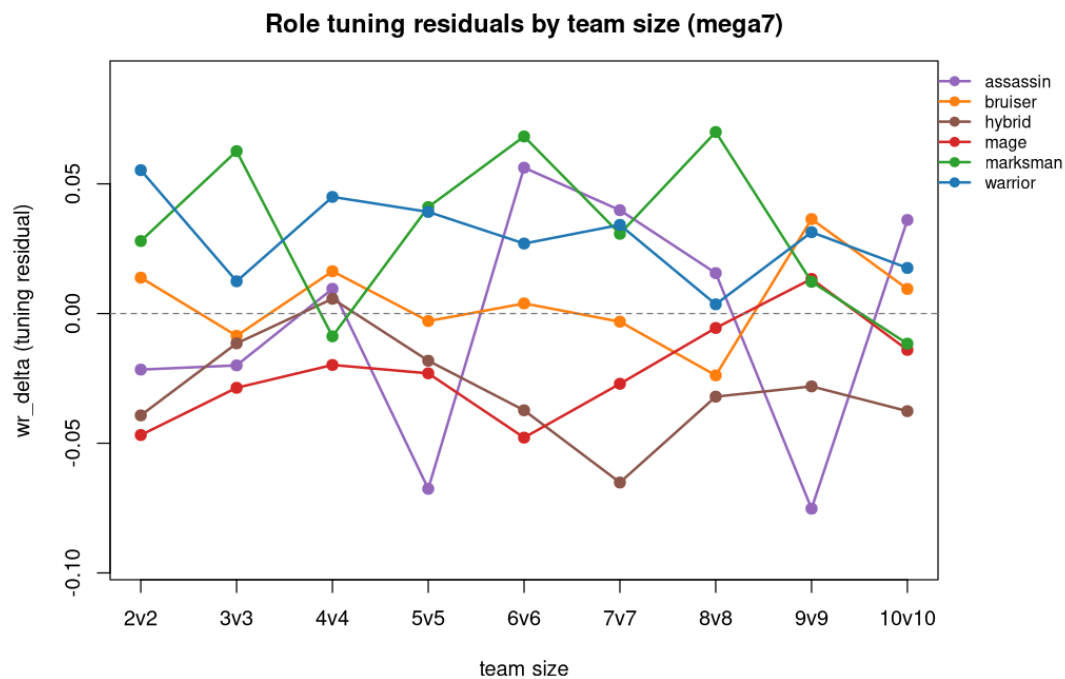
Mage is the floor at every size except 9v9. Hybrid and bruiser are the ceiling — the same two outliers flagged in mega6.

3.2 Tuning residuals (wr_delta)

Pooled (combined) role `wr_delta`:

role	<code>wr_delta</code>
hybrid	−0.030
mage	−0.018
assassin	−0.001
bruiser	+0.008
warrior	+0.024
marksman	+0.029

The story splits from raw win rate. ****Hybrid is the most *under-budget* role despite the highest raw win rate — its members are high-power T10 bosses that win on stats but under-deliver on kit.** Marksman and warrior over-deliver** (+0.025–0.029): they punch above their power budget. Mage is under on both raw and budget.

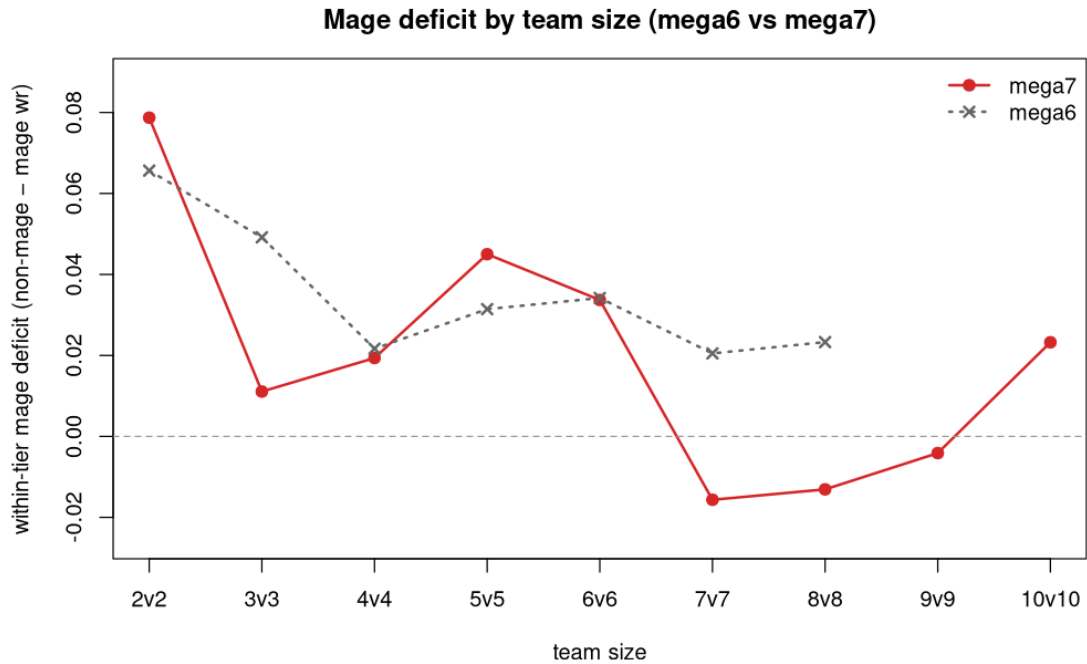


Role tuning residuals by team size

4. Mage deficit

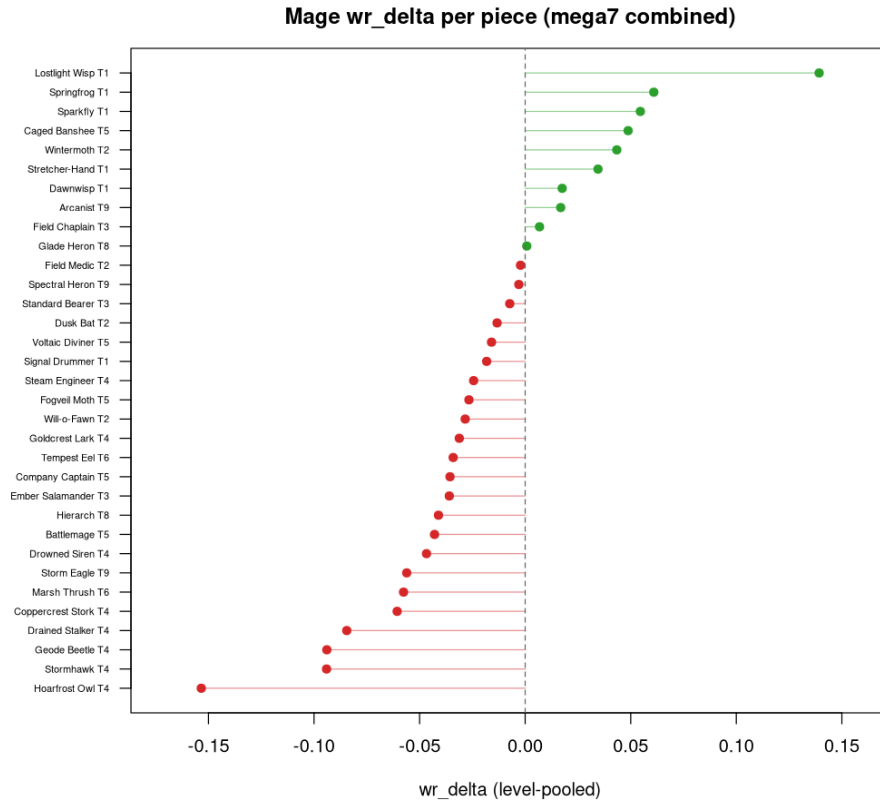
stage	mage	nonmage	within_tier_def	cor(tier,wr)
2v2	0.398	0.542	0.079	0.383
3v3	0.444	0.519	0.011	0.290
4v4	0.434	0.519	0.019	0.322
5v5	0.432	0.514	0.045	0.255
6v6	0.454	0.521	0.034	0.154
7v7	0.453	0.506	−0.016	0.265
8v8	0.480	0.511	−0.013	0.204
9v9	0.499	0.502	−0.004	0.223
10v10	0.452	0.510	0.023	0.223

The within-tier deficit is sharply concentrated at **2v2 (0.079)** and is gone or inverted by 7v7–8v8. Mages scale *into* relevance with team size — they need allies to survive long enough to ramp. Versus mega6 at matched sizes the picture is mixed: 2v2 slightly worse (+0.013), but 3v3/7v7/8v8 notably better (−0.036 to −0.038).



Mage deficit by team size: mega6 vs mega7

The worst individual mages (pooled `wr_delta`) are **Hierarch** (clear T8 L3, −0.221), **Hoarfrost Owl** (snow T4 L3, −0.184), **Standard Bearer** (clear T3 L2, −0.179), **Fogveil Moth** (mist T5 L3, −0.170), **Drowned Siren** (rain T4 L1, −0.165). Note Hierarch — it received the on-death barrier this session, which does nothing for its own survival; it remains the single most under-tuned mage. **Glade Heron L3 is on-budget (+0.022)** despite a 0.903 raw win rate (T8 L3 is *expected* to win that much) — the rework landed in the right place.

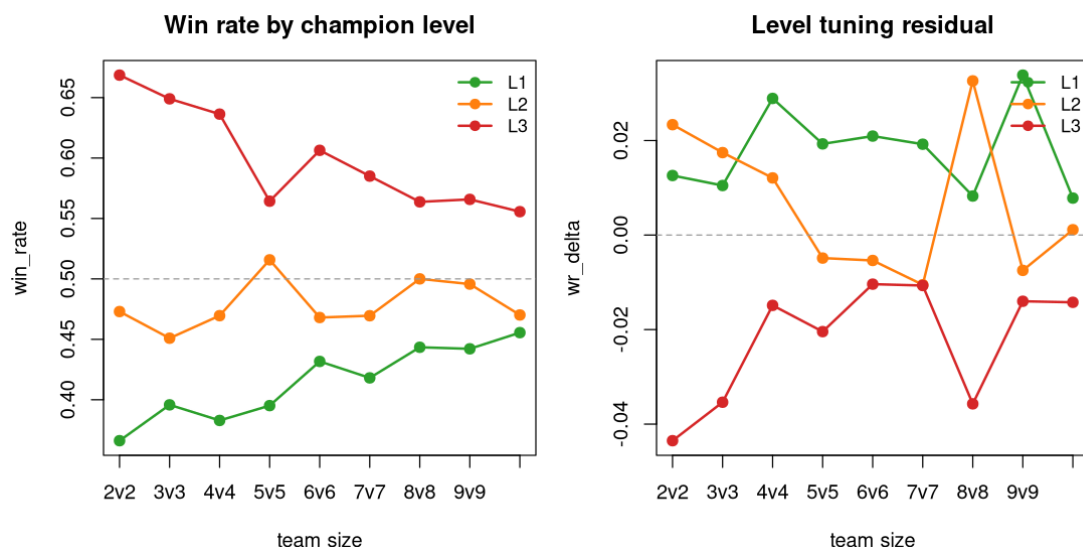


Mage wr\{\}_delta per piece (combined)

5. Level effects

level	win_rate	wr_delta	timeout
1	0.424	+0.019	0.279
2	0.480	+0.004	0.277
3	0.593	-0.021	0.304

The level premium is steep (L1→L3 = +0.17 raw) but **wr_delta** stays small: L1 slightly under-delivers, L3 slightly over. Mixed-level teams will be L3-dominated, which is by design. No action needed.



Win rate and tuning residual by champion level

6. Weather effects

Headline correction. A first pass on this dataset reported a dominant **+0.18** own-vs-counter weather advantage. That number was wrong — a metric artefact, not a balance fact. This section uses the corrected computation (raw per-weather win rates); the favor effect is real but **modest** ($\sim +0.01$).

The bug. The `counter_weather_wr` column defaults to **0.0** when a piece has no games in its counter weather. That conflates "no data" with "0% win rate". It fires constantly: a piece's counter weather is one specific weather of six, and every `clear`-affinity piece has **no** counter weather at all (NEUTRAL on the ring). Result — `counter_weather_wr == 0.0` in **37.1%** of rows. Averaging those zeros in collapses the mean counter and manufactures a fake gap:

computation	own	counter	own — counter
buggy column-average (zeros in)	0.505	0.332	+0.173
correct (raw per-weather win rate)	0.509	0.499	+0.010

Fixed in code: `weather_metrics` now emits `NaN` (not 0.0) for a weather the piece never played, and the CSV writer leaves the cell empty so aggregators skip it. This report recomputes from raw per-weather win rates and is unaffected.

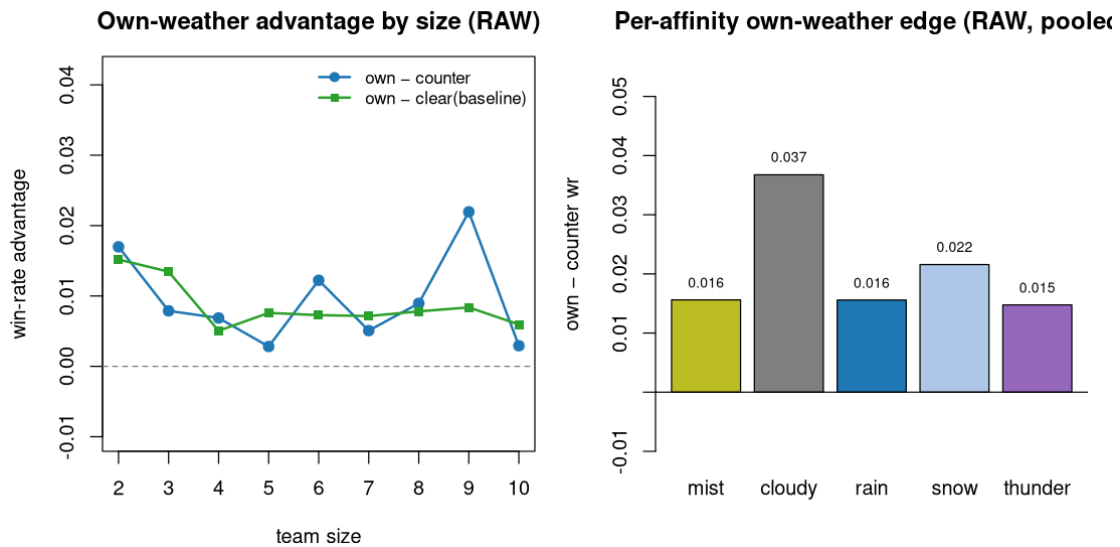
Corrected own-weather advantage by size (own = win rate in own affinity weather; clear = no-favor baseline; counter = weather that preys on you):

stage	clear (base)	own	counter	own — counter	own — clear
2v2	0.500	0.515	0.498	+0.017	+0.015
4v4	0.500	0.505	0.498	+0.007	+0.005
6v6	0.500	0.507	0.495	+0.012	+0.007
8v8	0.500	0.508	0.499	+0.009	+0.008
10v10	0.500	0.506	0.503	+0.003	+0.006

Corrected per-affinity (raw, pooled all stages):

affinity	clear	own	counter	own — clear	own — counter
mist	0.512	0.513	0.498	+0.002	+0.016
cloudy	0.506	0.530	0.494	+0.025	+0.037
rain	0.515	0.530	0.515	+0.016	+0.016
snow	0.509	0.516	0.495	+0.007	+0.022
thunder	0.491	0.511	0.496	+0.020	+0.015

Every affinity gains in its own weather and loses (or is flat) in its counter weather — **the favor system works as designed, and the earlier snow/thunder “inversion” was thin-slice noise**, not a real effect. The magnitude is small (single-digit win-rate points), so weather favor is a flavour-and-edge system, *not* a dominant axis — it does not overshadow kit or role. Whether ± 0.30 favor magnitude should produce a *larger* effect than this is a separate design question, but it is not overpowered.



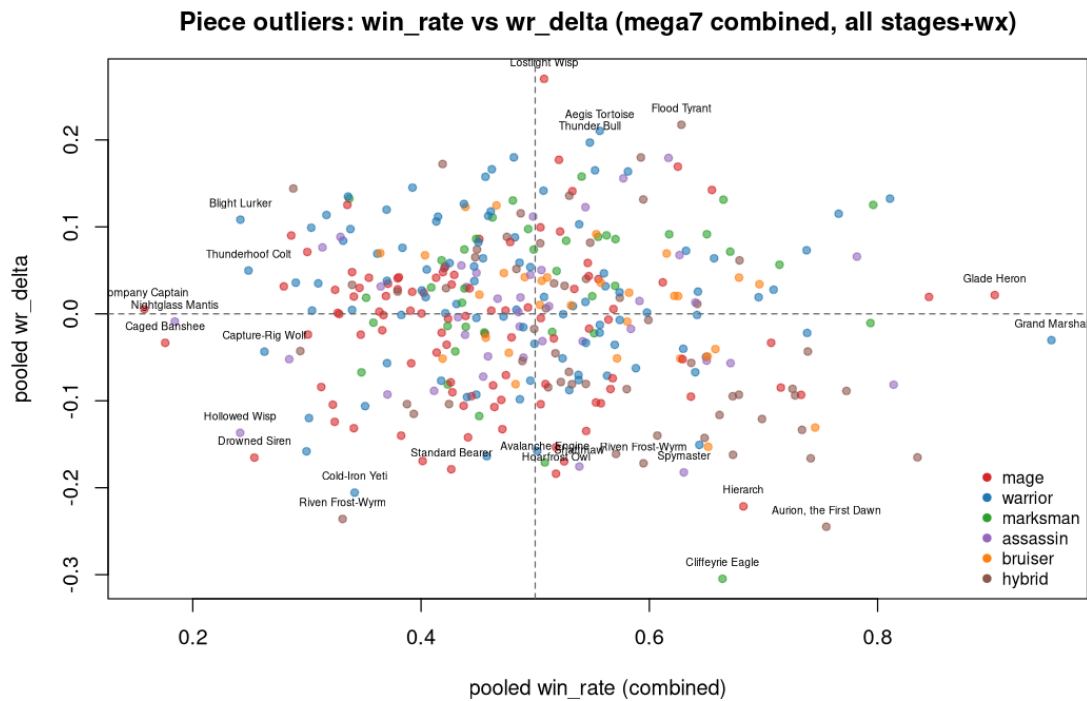
Weather: corrected own-advantage by size and per-affinity edge

7. Timeouts

role	2v2	5v5	8v8	10v10
assassin	0.131	0.312	0.352	0.483
bruiser	0.320	0.378	0.508	0.398
hybrid	0.120	0.289	0.350	0.405
mage	0.160	0.248	0.406	0.448
marksman	0.078	0.257	0.379	0.404
warrior	0.145	0.271	0.412	0.469

Bruisers stall hardest at small/mid sizes (high survivability, low burst), consistent with mega6. At the largest sizes nearly half of all battles time out — the 12,000-tick cap is too tight for 10v10 to resolve organically. Large-format role reads should be treated cautiously because of this.

8. Outliers and tuning priorities



Most under-tuned (pooled wr_delta):

name	affinity	role	tier	level	win_rate	wr_delta
Cliffeyrie Eagle	cloudy	marksman	9	3	0.664	-0.305
Aurion, the First Dawn	clear	hybrid	10	3	0.755	-0.245
Riven Frost-Wyrm	snow	hybrid	9	1	0.331	-0.236
Hierarch	clear	mage	8	3	0.682	-0.221
Cold-Iron Yeti	snow	warrior	4	2	0.342	-0.206
Hoarfrost Owl	snow	mage	4	3	0.518	-0.184
Spymaster	clear	assassin	8	3	0.630	-0.182

Most over-tuned:

name	affinity	role	tier	level	win_rate	wr_delta
Lostlight Wisp	mist	mage	1	1	0.508	+0.270
Flood Tyrant	rain	hybrid	10	2	0.628	+0.217
Aegis Tortoise	clear	warrior	5	3	0.557	+0.210
Thunder Bull	thunder	warrior	7	2	0.548	+0.197
Frostfang Wolverine	snow	warrior	8	1	0.481	+0.180

The under-tuned list is dominated by **high-tier hybrids/mages that win on raw power but miss their budget** (Aurion, Hierarch) and **snow pieces** (Riven Frost-Wyrm, Cold-Iron Yeti, Hoarfrost Owl) — reinforcing the snow-affinity anomaly from §6. The over-tuned list is led by a T1 mage (Lostlight Wisp) and the rain/thunder hybrid Tyrants.

9. Recommendations

- **Mage early-game power (priority).** Target the 2v2 within-tier deficit specifically — it vanishes at scale, so the fix is early survivability/damage, not late-game. Worst offenders: Hierarchy, Hoarfrost Owl, Standard Bearer, Fogveil Moth, Drowned Siren.
- **T10 hybrid boss kits (Aurion, Aerion, Storm Tyrant) under-deliver vs budget** — recurring since mega6. Their kits need a pass; they should top the chart, not merely win on raw power.
- **Hierarchy.** Most under-tuned mage; the new on-death barrier helps allies, not itself. Needs a personal-survivability or damage buff.
- **Weather favor: no action needed.** Corrected analysis (§6) shows it working as designed at a modest $\sim +0.01$ edge. If a **stronger** competitive weather effect is wanted, the ± 0.30 magnitude could be revisited — but it is not a balance problem. The metric bug that overstated it has been fixed in `weather_metrics`.
- **Validated this session:** Glade Heron L3 is on-budget ($+0.022$); the poison percentage-decay model produced no runaway poison outliers in team play.

10. Reproducibility

```
# 1. Run the sweep (current working tree)
python -m tools.simulation.mega --total-battles <N> --skip 1v1 \
    --workers 16 --out results/mega/mega7

# 2. Analysis + tables + cache
Rscript reviews/mega_sim/11_mega7.R

# 3. Figures
Rscript reviews/mega_sim/12_mega7_plots.R

# 4. This PDF
python3 reviews/mega_sim/build_mega7_pdf.py
```

Tables written to `reviews/mega_sim/tables/m7_*.csv`; cache to `cache_mega7.rds`; figures to `reviews/mega_sim/plots/m7_*.png`. The engine is byte-deterministic — same tree + same flags reproduce the CSVs exactly.